

Homework #1. Due on Thursday, Sep 3rd, 11:59pm on Canvas

Plan for next week (Aug 31, Sep 2):

More examples of vector spaces, subspaces, spans, linear independences, bases and dimension (sections 1.2-1.5 and parts of 1.6 of the book Linear Algebra by Friedberg, Insel and Spence).

Notations: $\mathbb{N} = \{1, 2, \dots\}$ denotes the natural numbers. Given a field F and $n, m \in \mathbb{N}$, the set of $n \times m$ matrices with entries in F is denoted by $\text{Mat}_{n \times m}(F)$. The notation $\text{Mat}_n(F)$ is a shortcut for $\text{Mat}_{n \times n}(F)$.

Problems:

Problem 1: Let F be a field, and let $A \in \text{Mat}_{n \times m}(F)$ and $B \in \text{Mat}_{m \times k}(F)$ for some $n, m, k \in \mathbb{N}$. Prove that $(AB)^T = B^T A^T$ where X^T is the transpose of a matrix X . Include all the details.

Problem 2: Let F be a field and $n \in \mathbb{N}$. Given $A = (a_{ij}) \in \text{Mat}_n(F)$, the *trace* of A , denoted by $\text{tr}(A)$, is the sum of all diagonal entries of A :

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

- (a) Prove that $\text{tr}(\lambda A + \mu B) = \lambda \text{tr}(A) + \mu \text{tr}(B)$ for all $A, B \in \text{Mat}_n(F)$ and $\lambda, \mu \in F$.
- (b) Let $\mathfrak{sl}_n(F) = \{A \in \text{Mat}_n(F) : \text{tr}(A) = 0\}$. Prove that $\mathfrak{sl}_n(F)$ is a subspace of $\text{Mat}_n(F)$ (where $\text{Mat}_n(F)$ is considered as a vector space over F with respect to the usual matrix addition and scalar multiplication).
- (c) Prove that $\text{tr}(AB) = \text{tr}(BA)$ for all $A, B \in \text{Mat}_{n \times n}(F)$.

Problem 3: Let F be a field. In both parts below $V = F^2 = \{(a, b) : a, b \in F\}$ as a set. In each part determine (with proof) whether V is a vector space with respect to given operations. If your answer is yes, find an explicit formula for the zero element of V (note that the zero element need not equal $(0, 0)$):

- (a) addition is defined in the usual way: $(a, b) + (c, d) = (a + c, b + d)$, and scalar multiplication is defined by $\lambda(a, b) = (\lambda a, b)$ for all $\lambda \in F$ and $(a, b) \in V$;
- (b) addition is defined by $(a, b) + (c, d) = (a + c, b + d + 1)$, and scalar multiplication is defined by $\lambda(a, b) = (\lambda a, \lambda b + \lambda - 1)$

Problem 4: Let V be the vector space of all polynomials with real coefficients (that is, all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ of the form $f(x) = \sum_{i=0}^n a_i x^i$ for some $n \in \mathbb{N}$ and $a_0, \dots, a_n \in \mathbb{R}$). For each of the following subsets W of V determine (with proof) if W is a subspace of V :

- (a) $W = \{f \in V : f(1) \geq 0\}$;
- (b) $W = \{f \in V : f(1) = f(2)\}$.

Problem 5: Let F be a field, $n \in \mathbb{N}$ and $V = F^n = \{(x_1, \dots, x_n) : x_i \in F \text{ for each } i\}$. We view V as a vector space with respect to the usual addition and scalar multiplication. Let $W = \{(x_1, \dots, x_n) \in V : \sum_{i=1}^n x_i = 0\}$.

- (a) Prove that W is a subspace of V .
- (b) Now assume F is finite. Find $|W|$ (with proof). The answer should be expressed in terms of $|F|$ and n . **Note:** As explained in Lecture 1, basic examples of finite fields are $\mathbb{Z}_p$ (integers mod p) for any prime p , but not every finite field is of this form (even up to isomorphism).

Problem 6: Let V be a vector space over a field F . Let W_1 and W_2 be subspaces of V .

- (a) Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
- (b) Prove that $W_1 + W_2$ is a subspace of V , where by definition $W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$.
- (c) Prove that $W_1 + W_2$ is the smallest subspace of V containing W_1 and W_2 , that is, if W is any subspace of V containing W_1 and W_2 , then W contains $W_1 + W_2$.

Problem 7: Let V be a vector space over $\mathbb{C}$ (complex numbers).

- (a) Explain why V is also a vector space over $\mathbb{R}$ (reals) and $\mathbb{Q}$ (rationals) with respect to the same operations.

According to (a), given a subset W of V , we can ask whether it is a $\mathbb{C}$ -subspace (that is, a subspace of V considered as a vector space over $\mathbb{C}$), and $\mathbb{R}$ -subspace or a $\mathbb{Q}$ -subspace. These are 3 different notions. For example, it is easier to be an $\mathbb{Q}$ -subspace than an $\mathbb{R}$ -subspace since to be a $\mathbb{Q}$ -subspace W only needs to be closed under addition and multiplications by rationals, while to be a $\mathbb{R}$ -subspace, it needs to be also closed under multiplications by all reals.

In parts (b) and (c) $V = \mathbb{C}$ (where the vector space addition is the addition in $\mathbb{C}$ and the scalar multiplication is the usual multiplication).

- (a) Prove that the only $\mathbb{C}$ -subspaces of V are V itself and $\{0\}$.

- (b) Describe all $\mathbb{R}$ -subspaces of V (and prove that there are no other subspaces). You may use any results from the Chapter 1 of the book, but state your references clearly.

Problem 8: Let V be a vector space over a field F and $S = \{v_1, v_2\}$ a subset of V containing precisely two elements. Prove that S is linearly dependent $\iff v_1$ is a scalar multiple of v_2 or v_2 is a scalar multiple of v_1 . (By definition, w is a scalar multiple of v if $w = \lambda v$ for some $\lambda \in F$).